

RISC-V ISA Notes

What is RISC-V?

RISC-V is an open-source Instruction Set Architecture (ISA) designed using Reduced Instruction Set Computing principles.

Features

- Open Source
 - Modular
 - Simple instruction format
 - Easy to implement
 - Suitable for embedded systems
-

RISC vs CISC

RISC	CISC
Simple instructions	Complex instructions
Fixed instruction length	Variable instruction length
Faster execution	More complex decoding
Example: RISC-V	Example: x86

Register File

RISC-V contains 32 registers.

Examples:

x0 → Always 0

x1 → Return Address

x2 → Stack Pointer

x5-x31 → General Purpose Registers

Instruction Formats

R-Type

Used for:

- ADD

- SUB
 - AND
 - OR
 - XOR
-

I-Type

Used for:

- ADDI
 - LW
-

S-Type

Used for:

- SW
-

B-Type

Used for:

- BEQ
 - BNE
-

J-Type

Used for:

- JAL
-

Arithmetic Instructions

ADD

add x3,x1,x2

SUB

sub x3,x1,x2

ADDI

addi x1,x0,5

Logical Instructions

AND

OR

XOR

Memory Instructions

LW

Load data from memory.

SW

Store data into memory.

Branch Instructions

BEQ

Jump if equal.

BNE

Jump if not equal.

Jump Instructions

JAL

Jump and Link.

Key Learning

- Learned ISA concepts
- Practiced assembly programming
- Understood register architecture
- Learned instruction formats